

Лабораторная работа №15

Управление логическими томами

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Цель работы

Получить навыки работы с физическими томами, группами томов и логическими томами в LVM.

Ход выполнения

Создание раздела /dev/sdb1

```
Welcome to fdisk (util-linux 2.40.2).
```

```
Changes will remain in memory only, until you decide to write them.  
Be careful before using the write command.
```

```
Command (m for help): n
```

```
Partition type
```

- p primary (0 primary, 0 extended, 4 free)
- e extended (container for logical partitions)

```
Select (default p): p
```

```
Partition number (1-4, default 1):
```

```
First sector (2048-3145727, default 2048):
```

```
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-3145727, default 3145727): +300M
```

```
Created a new partition 1 of type 'Linux' and of size 300 MiB.
```

```
Command (m for help): t
```

```
Selected partition 1
```

```
Hex code or alias (type L to list all): 8e
```

```
Changed type of partition 'Linux' to 'Linux LVM'.
```

```
Command (m for help): w
```

```
The partition table has been altered.
```

```
Calling ioctl() to re-read partition table.
```

```
Syncing disks.
```

```
root@amnauruzova:/home/amnauruzova# partprobe /dev/sdb
```

```
root@amnauruzova:/home/amnauruzova# pvcreate /dev/sdb1
```

```
Physical volume "/dev/sdb1" successfully created.
```

```
root@amnauruzova:/home/amnauruzova# pvs
```

PV	VG	Fmt	Attr	PSize	PFree
/dev/sda3	rl_vbox	lvm2	a--	<39.00g	0
/dev/sdb1		lvm2	---	300.00m	300.00m

```
root@amnauruzova:/home/amnauruzova# █
```

Создание физического тома

```
root@amnauruzova: /home/amnauruzova#  
root@amnauruzova: /home/amnauruzova# pvs  
PV          VG      Fmt Attr PSize  PFree  
/dev/sda3   rl_vbox lvm2 a-- <39.00g  0  
/dev/sdb1   lvm2 --- 300.00m 300.00m  
root@amnauruzova: /home/amnauruzova# vgcreate vgdata /dev/sdb1  
Volume group "vgdata" successfully created  
root@amnauruzova: /home/amnauruzova# vgs  
VG      #PV #LV #SN Attr   VSize  VFree  
rl_vbox 1  2  0 wz--n- <39.00g  0  
vgdata  1  0  0 wz--n- 296.00m 296.00m  
root@amnauruzova: /home/amnauruzova# pvs  
PV          VG      Fmt Attr PSize  PFree  
/dev/sda3   rl_vbox lvm2 a-- <39.00g  0  
/dev/sdb1   vgdata  lvm2 a-- 296.00m 296.00m  
root@amnauruzova: /home/amnauruzova# lvcreate -n lvdata -l 50%FREE vgdata  
Logical volume "lvdata" created.  
root@amnauruzova: /home/amnauruzova# lvs  
LV      VG      Attr      LSize  Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert  
root    rl_vbox -wi-ao---- 35.05g  
swap    rl_vbox -wi-ao---- <3.95g  
lvdata  vgdata  -wi-a----- 148.00m  
root@amnauruzova: /home/amnauruzova#
```

Рис. 2: Создание PV

Создание группы томов и логического тома

```
root@amnauruzova: /home/amnauruzova#  
root@amnauruzova:/home/amnauruzova# mkfs.ext4 /dev/vgdata/lvdata  
mke2fs 1.47.1 (20-May-2024)  
Creating filesystem with 151552 1k blocks and 37848 inodes  
Filesystem UUID: 648b93ea-6eda-45dc-84e7-e9b3e699f4ad  
Superblock backups stored on blocks:  
        8193, 24577, 40961, 57345, 73729  
  
Allocating group tables: done  
Writing inode tables: done  
Creating journal (4096 blocks): done  
Writing superblocks and filesystem accounting information: done  
  
root@amnauruzova:/home/amnauruzova# mkdir -p /mnt/data  
root@amnauruzova:/home/amnauruzova#
```

Рис. 3: Создание VG и LV

Добавление точки монтирования

```
GNU nano 8.1 /etc/fstab

#
# /etc/fstab
# Created by anaconda on Tue Sep  2 12:53:30 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=34884054-27f2-4624-8671-1f66bf893ae2 /                    xfs     defaults        0 0
UUID=5bb25999-de96-4a24-930c-75854bf11fc1 /boot          xfs     defaults        0 0
UUID=774abe3d-c83f-4c4f-90c6-404f43a900ee none           swap    defaults        0 0
/dev/vgdata/lvdata /mnt/data      ext4    defaults        1 2

#UUID=e97bbe69-dd87-4579-880c-1ca10bbbf317 /mnt/data      xfs     defaults        1 2
#UUID=ef6eb4a8-920f-4d8b-96f8-b4e9a925c30b /mnt/data-ext  ext4    defaults        1 2
#UUID=6c12fcce-02a4-4084-8214-c952a4d0cbda none           swap    defaults        0 0
```

Рис. 4: fstab


```
root@amnauruzova:/home/amnauruzova#  
root@amnauruzova:/home/amnauruzova# mount -a  
mount: (hint) your fstab has been modified, but systemd still uses  
the old version; use 'systemctl daemon-reload' to reload.  
root@amnauruzova:/home/amnauruzova# mount | grep mnt  
/dev/mapper/vgdata-lvdata on /mnt/data type ext4 (rw,relatime,seclabel)  
root@amnauruzova:/home/amnauruzova#
```

Рис. 5: Проверка монтирования

Создание нового раздела /dev/sdb2

```
Command (m for help): t
Partition number (1,2, default 2):
Hex code or alias (type L to list all): 8e

Changed type of partition 'Linux' to 'Linux LVM'.

Command (m for help): p
Disk /dev/sdb: 1.5 GiB, 1610612736 bytes, 3145728 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcb7b5831
```

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sdb1		2048	616447	614400	300M	8e	Linux LVM
/dev/sdb2		616448	1230847	614400	300M	8e	Linux LVM

```
Filesystem/RAID signature on partition 2 will be wiped.

Command (m for help): w
The partition table has been altered.
Syncing disks.

root@amnauruzova:/home/amnauruzova#
```

Расширение группы томов vgdata

```
root@amnauruzova:/home/amnauruzova#  
root@amnauruzova:/home/amnauruzova# pvcreate /dev/sdb2  
Physical volume "/dev/sdb2" successfully created.  
root@amnauruzova:/home/amnauruzova# vgextend vgdata /dev/sdb2  
Volume group "vgdata" successfully extended  
root@amnauruzova:/home/amnauruzova# vgs  
VG      #PV #LV #SN Attr   VSize  VFree  
rl_vbox 1  2  0 wz--n- <39.00g    0  
vgdata  2  1  0 wz--n- 592.00m 444.00m  
root@amnauruzova:/home/amnauruzova# lvs  
LV      VG      Attr   LSize   Pool Origin Data%  Meta%   Move Log Cpy%Sync Convert  
root   rl_vbox -wi-ao---- 35.05g  
swap   rl_vbox -wi-ao---- <3.95g  
lvdata vgdata -wi-ao---- 148.00m  
root@amnauruzova:/home/amnauruzova# df -h  
Filesystem                Size  Used Avail Use% Mounted on  
/dev/mapper/rl_vbox-root    35G   6.1G   29G   18% /  
devtmpfs                   4.0M    0   4.0M    0% /dev  
tmpfs                      1.8G   84K   1.8G    1% /dev/shm  
tmpfs                      731M   9.3M   722M    2% /run  
tmpfs                      1.0M    0   1.0M    0% /run/credentials/systemd-journald.service  
/dev/sda2                  960M  377M   584M   40% /boot  
tmpfs                      366M  144K   366M    1% /run/user/1000  
tmpfs                      366M   60K   366M    1% /run/user/0  
/dev/mapper/vgdata-lvdata  134M   14K   123M    1% /mnt/data  
root@amnauruzova:/home/amnauruzova#
```

Рис. 7: Расширение VG

Увеличение логического тома lvdata

```
root@amnauruzova:/home/amnauruzova#
root@amnauruzova:/home/amnauruzova# lvextend -r -l +50%FREE /dev/vgdata/lvdata
File system ext4 found on vgdata/lvdata mounted at /mnt/data.
Size of logical volume vgdata/lvdata changed from 148.00 MiB (37 extents) to 372.00 MiB (93 extents).
Extending file system ext4 to 372.00 MiB (390070272 bytes) on vgdata/lvdata...
resize2fs /dev/vgdata/lvdata
resize2fs 1.47.1 (20-May-2024)
Filesystem at /dev/vgdata/lvdata is mounted on /mnt/data; on-line resizing required
old_desc_blocks = 2, new_desc_blocks = 3
The filesystem on /dev/vgdata/lvdata is now 380928 (1k) blocks long.

resize2fs done
Extended file system ext4 on vgdata/lvdata.
Logical volume vgdata/lvdata successfully resized.
root@amnauruzova:/home/amnauruzova# lvs
LV      VG      Attr      LSize   Pool Origin Data%  Meta%   Move Log Cpy%Sync Convert
root    rl_vbox -wi-ao---- 35.05g
swap    rl_vbox -wi-ao----  <3.95g
lvdata   vgdata  -wi-ao---- 372.00m
root@amnauruzova:/home/amnauruzova# df -h
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/rl_vbox-root    35G        6.1G   29G  18% /
devtmpfs                   4.0M         0   4.0M   0% /dev
tmpfs                      1.8G       84K   1.8G   1% /dev/shm
tmpfs                      731M       9.3M   722M   2% /run
tmpfs                      1.0M         0   1.0M   0% /run/credentials/systemd-journald.service
/dev/sda2                  960M      377M   584M  40% /boot
tmpfs                      366M      144K   366M   1% /run/user/1000
tmpfs                      366M       60K   366M   1% /run/user/0
/dev/mapper/vgdata-lvdata  344M       14K   324M   1% /mnt/data
root@amnauruzova:/home/amnauruzova#
```

Рис. 8: Увеличение LV

Уменьшение логического тома

```
resize2fs done
remount /dev/vgdata/lvdata /mnt/data
mount: (hint) your fstab has been modified, but systemd still uses
       the old version; use 'systemctl daemon-reload' to reload.
remount done
Reduced file system ext4 on vgdata/lvdata.
Size of logical volume vgdata/lvdata changed from 372.00 MiB (93 extents) to 324.00 MiB (81 extents).
Logical volume vgdata/lvdata successfully resized.
root@amnauruzova:/home/amnauruzova# lvs
LV      VG      Attr      LSize   Pool Origin Data%  Meta%   Move Log Cpy%Sync Convert
root    rl_vbox -wi-ao---- 35.05g
swap    rl_vbox -wi-ao---- <3.95g
lvdata  vgdata  -wi-ao---- 324.00m
root@amnauruzova:/home/amnauruzova# df -h
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/rl_vbox-root  35G        6.1G   29G   18% /
devtmpfs                   4.0M         0   4.0M    0% /dev
tmpfs                      1.8G       84K   1.8G    1% /dev/shm
tmpfs                      731M       9.3M   722M    2% /run
tmpfs                      1.0M         0   1.0M    0% /run/credentials/systemd-journald.service
/dev/sda2                  960M      377M   584M   40% /boot
tmpfs                      366M      144K   366M    1% /run/user/1000
tmpfs                      366M       60K   366M    1% /run/user/0
/dev/mapper/vgdata-lvdata 298M       14K   280M    1% /mnt/data
root@amnauruzova:/home/amnauruzova#
```

Рис. 9: Уменьшение LV

Самостоятельная работа

Создание раздела на диске /dev/sdc

```
First sector (2048-3145727, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-3145727, default 3145727): +400M

Created a new partition 1 of type 'Linux' and of size 400 MiB.
Partition #1 contains a xfs signature.

Do you want to remove the signature? [Y]es/[N]o: Y

The signature will be removed by a write command.

Command (m for help): t
Selected partition 1
Hex code or alias (type L to list all): 8e
Changed type of partition 'Linux' to 'Linux LVM'.

Command (m for help): p
Disk /dev/sdc: 1.5 GiB, 1610612736 bytes, 3145728 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd4a244f4

Device      Boot Start    End Sectors  Size Id Type
/dev/sdc1           2048 821247  819200    400M 8e Linux LVM

Filesystem/RAID signature on partition 1 will be wiped.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

```
root@amnapautuzova: /home/amnapautuzova#
```

Создание PV, VG и LV

```
root@amnauruzova:/home/amnauruzova#
root@amnauruzova:/home/amnauruzova# pvcreate /dev/sdc1
Physical volume "/dev/sdc1" successfully created.
root@amnauruzova:/home/amnauruzova# vgcreate vgggroup /dev/sdc1
Volume group "vgggroup" successfully created
root@amnauruzova:/home/amnauruzova# lvcreate -n lvgroup -l 100%FREE vgggroup
Logical volume "lvgroup" created.
root@amnauruzova:/home/amnauruzova# pvs
PV          VG          Fmt Attr PSize  PFree
/dev/sda3   rl_vbox     lvm2 a--  <39.00g    0
/dev/sdb1   vgdata     lvm2 a--  296.00m    0
/dev/sdb2   vgdata     lvm2 a--  296.00m 268.00m
/dev/sdc1   vgggroup    lvm2 a--  396.00m    0
root@amnauruzova:/home/amnauruzova# vgs
VG          #PV #LV #SN Attr   VSize  VFree
rl_vbox     1   2   0 wz--n-  <39.00g    0
vgdata      2   1   0 wz--n- 592.00m 268.00m
vgggroup     1   1   0 wz--n- 396.00m    0
root@amnauruzova:/home/amnauruzova# lvs
LV          VG          Attr      LSize  Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
root        rl_vbox     -wi-ao---- 35.05g
swap        rl_vbox     -wi-ao----  <3.95g
lvdata      vgdata      -wi-ao---- 324.00m
lvgroup     vgggroup    -wi-a----- 396.00m
root@amnauruzova:/home/amnauruzova#
```

Рис. 11: Создание PV, VG и LV


```
root@amnauruzova:/home/amnauruzova#  
root@amnauruzova:/home/amnauruzova# mkdir -p /mnt/groups  
root@amnauruzova:/home/amnauruzova# mkfs.xfs /dev/vggroup/lvgroup  
meta-data=/dev/vggroup/lvgroup  isize=512    agcount=4, agsize=25344 blks  
        =                       sectsz=512   attr=2, projid32bit=1  
        =                       crc=1        finobt=1, sparse=1, rmapbt=1  
        =                       reflink=1    bigtime=1 inobtcount=1 nnext64=1  
        =                       exchange=0  
data      =                       bsize=4096   blocks=101376, imaxpct=25  
        =                       sunit=0      swidth=0 blks  
naming    =version 2             bsize=4096   ascii-ci=0, ftype=1, parent=0  
log       =internal log        bsize=4096   blocks=16384, version=2  
        =                       sectsz=512   sunit=0 blks, lazy-count=1  
realtime  =none                 extsz=4096    blocks=0, rtextents=0  
root@amnauruzova:/home/amnauruzova#
```

Рис. 12: Форматирование XFS

Добавление точки монтирования

```
GNU nano 8.1 /etc/fstab

#
# /etc/fstab
# Created by anaconda on Tue Sep  2 12:53:30 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=34884054-27f2-4624-8671-1f66bf893ae2 / xfs defaults 0 0
UUID=5bb25999-de96-4a24-930c-75854bf11fc1 /boot xfs defaults 0 0
UUID=774abe3d-c83f-4c4f-90c6-404f43a900ee none swap defaults 0 0
/dev/vgdata/lvdata /mnt/data ext4 defaults 1 2
/dev/vggroup/lvgroup /mnt/groups xfs defaults 1 2

#UUID=e97bbe69-dd87-4579-880c-1ca10bbb317 /mnt/data xfs defaults 1 2
#UUID=ef6eb4a8-920f-4d8b-96f8-b4e9a925c30b /mnt/data-ext ext4 defaults 1 2
#UUID=6c12fcce-02a4-4084-8214-c952a4d0cbda none swap defaults 0 0
```

Рис. 13: fstab

Проверка подключения файловой системы

```
root@amnauruzova: /home/amnauruzova#  
root@amnauruzova: /home/amnauruzova# mount | grep mnt  
/dev/mapper/vggroup-lvgroup on /mnt/groups type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota)  
/dev/mapper/vgdata-lvdata on /mnt/data type ext4 (rw,relatime,seclabel)  
root@amnauruzova: /home/amnauruzova# df -h  
Filesystem                Size  Used Avail Use% Mounted on  
/dev/mapper/rl_vbox-root    35G   6.1G   29G   18% /  
devtmpfs                   4.0M    0   4.0M    0% /dev  
tmpfs                       1.8G   84K   1.8G    1% /dev/shm  
tmpfs                       731M   9.3M   722M    2% /run  
tmpfs                       1.0M    0   1.0M    0% /run/credentials/systemd-journald.service  
/dev/sdb2                   960M  377M   584M   40% /boot  
/dev/mapper/vggroup-lvgroup 332M   27M   306M    8% /mnt/groups  
/dev/mapper/vgdata-lvdata   298M   14K   280M    1% /mnt/data  
tmpfs                       366M  140K   366M    1% /run/user/1000  
tmpfs                       366M   60K   366M    1% /run/user/0  
root@amnauruzova: /home/amnauruzova#
```

Рис. 14: Проверка монтирования

Создание нового раздела /dev/sda2

Created a new partition 2 of type 'Linux' and of size 300 MiB.

Command (m for help): t

Partition number (1,2, default 2):

Hex code or alias (type L to list all): 8e

Changed type of partition 'Linux' to 'Linux LVM'.

Command (m for help): p

Disk /dev/sda: 1.5 GiB, 1610612736 bytes, 3145728 sectors

Disk model: VBOX HARDDISK

Units: sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disklabel type: dos

Disk identifier: 0xd4a244f4

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sda1		2048	821247	819200	400M	8e	Linux LVM
/dev/sda2		821248	1435647	614400	300M	8e	Linux LVM

Command (m for help): w

The partition table has been altered.

Syncing disks.

root@amnauruzova:/home/amnauruzova#

Увеличение LV и файловой системы XFS

```
root@amnauruzova: /home/amnauruzova#
root@amnauruzova: /home/amnauruzova# pvcreate /dev/sda2
Physical volume "/dev/sda2" successfully created.
root@amnauruzova: /home/amnauruzova# vgextend vgggroup /dev/sda2
Volume group "vgggroup" successfully extended
root@amnauruzova: /home/amnauruzova# lvextend -r -l +100%FREE /dev/vgggroup/lvggroup
File system xfs found on vgggroup/lvggroup mounted at /mnt/groups.
Size of logical volume vgggroup/lvggroup changed from 396.00 MiB (99 extents) to 692.00 MiB (173 extents).
Extending file system xfs to 692.00 MiB (725614592 bytes) on vgggroup/lvggroup...
xfs_growfs /dev/vgggroup/lvggroup
meta-data=/dev/mapper/vgggroup-lvggroup isize=512    agcount=4, agsize=25344 blks
        =                               sectsz=512   attr=2, projid32bit=1
        =                               crc=1        finobt=1, sparse=1, rmapbt=1
        =                               reflink=1    bigtime=1 inobtcount=1 nnext64=1
        =                               exchange=0
data      =                               bsize=4096   blocks=101376, imaxpct=25
        =                               sunit=0      swidth=0 blks
naming    =version 2                   bsize=4096   ascii-ci=0, ftype=1, parent=0
log       =internal log                bsize=4096   blocks=16384, version=2
        =                               sectsz=512   sunit=0 blks, lazy-count=1
realtime  =none                       extsz=4096   blocks=0, rtextents=0
data blocks changed from 101376 to 177152
xfs_growfs done
Extended file system xfs on vgggroup/lvggroup.
Logical volume vgggroup/lvggroup successfully resized.
root@amnauruzova: /home/amnauruzova#
```

Рис. 16: Расширение LV и FS

Проверка PV, VG и LV

```
root@amnauruzova:/home/amnauruzova#
root@amnauruzova:/home/amnauruzova# pvs
PV          VG          Fmt Attr PSize  PFree
/dev/sda1   vggroup   lvm2 a-- 396.00m  0
/dev/sda2   vggroup   lvm2 a-- 296.00m  0
/dev/sdb3   rl_vbox   lvm2 a-- <39.00g  0
/dev/sdc1   vgdata    lvm2 a-- 296.00m  0
/dev/sdc2   vgdata    lvm2 a-- 296.00m 268.00m
root@amnauruzova:/home/amnauruzova# vgs
VG          #PV #LV #SN Attr   VSize  VFree
rl_vbox     1  2  0 wz--n- <39.00g  0
vgdata      2  1  0 wz--n- 592.00m 268.00m
vggroup     2  1  0 wz--n- 692.00m  0
root@amnauruzova:/home/amnauruzova# lvs
LV          VG          Attr      LSize   Pool Origin Data%  Meta%   Move Log Cpy%Sync Convert
root        rl_vbox     -wi-ao---- 35.05g
swap        rl_vbox     -wi-ao---- <3.95g
lvdata      vgdata      -wi-ao---- 324.00m
lvgroup     vggroup     -wi-ao---- 692.00m
root@amnauruzova:/home/amnauruzova# df -h
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/rl_vbox-root   35G        6.1G   29G   18% /
devtmpfs                   4.0M        0    4.0M    0% /dev
tmpfs                      1.8G       84K    1.8G    1% /dev/shm
tmpfs                      731M       9.3M   722M    2% /run
tmpfs                      1.0M        0     1.0M    0% /run/credentials/systemd-journald.service
/dev/sdb2                  960M       377M   584M   40% /boot
/dev/mapper/vggroup-lvgroup 628M       33M   596M    6% /mnt/groups
/dev/mapper/vgdata-lvdata   298M       14K   280M    1% /mnt/data
tmpfs                      366M      140K   366M    1% /run/user/1000
tmpfs                      366M       60K   366M    1% /run/user/0
root@amnauruzova:/home/amnauruzova#
```

Рис. 17: Проверка LV, VG, PV

Итоги работы

Получены практические навыки создания, расширения и уменьшения логических томов, а также управления файловыми системами EXT4 и XFS.